

Homework #3 - MATH 261A

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Problem 1

For an estimator to be unbiased, this means that the expectation of the estimator is equal to the parameter:

$$E(\vec{\mathbf{b}}) = \vec{\beta}$$

The least squares estimator for the coefficients, $\vec{\mathbf{b}}$, is equal to the following:

$$\vec{\mathbf{b}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$$

If we substitute the formula for linear regression model in for $\mathbf{Y} = \mathbf{X}\vec{\beta} + \vec{\epsilon}$ and simplify, we get:

$$\begin{aligned}\vec{\mathbf{b}} &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y} \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{X}\vec{\beta} + \vec{\epsilon}) \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} [\mathbf{X}^\top \mathbf{X}\vec{\beta} + \mathbf{X}^\top \vec{\epsilon}] \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}\vec{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \vec{\epsilon} \\ &= \vec{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \vec{\epsilon}\end{aligned}$$

Now if we apply the expectation operation, conditional on the data, $\mathbf{X}$, we get:

$$\begin{aligned}E(\vec{\mathbf{b}}) &= E(\vec{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \vec{\epsilon}) \\ &= E(\vec{\beta}) + E((\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \vec{\epsilon}) \\ &= \vec{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \cdot E(\vec{\epsilon}) \\ &= \vec{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \cdot \vec{\mathbf{0}} \\ &= \vec{\beta}\end{aligned}$$

Therefore, we have shown that our least squares estimator is unbiased if the model's assumptions are satisfied.

Problem 2

Part (a)

Part (b)

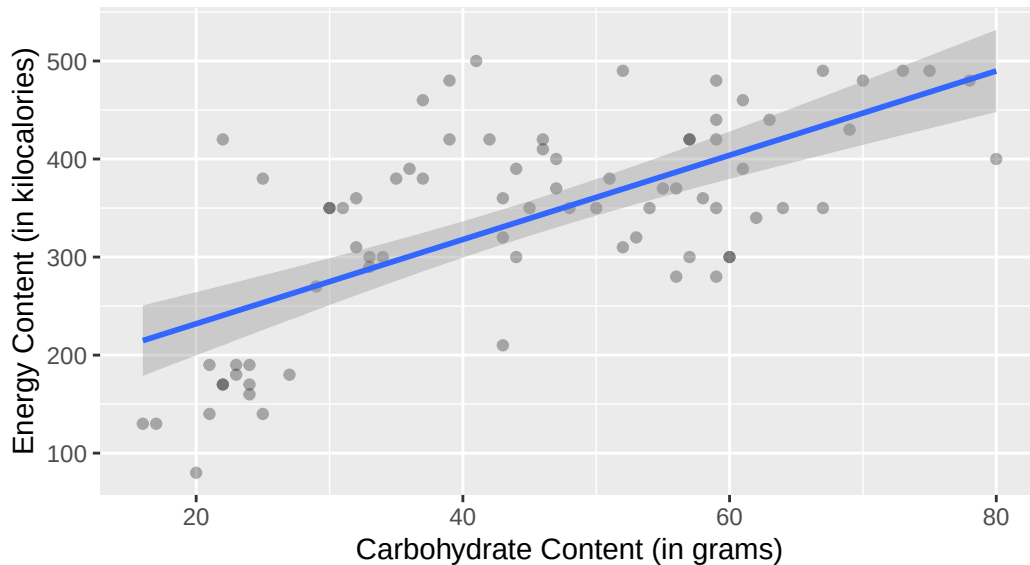
Problem 3

Problem 4

Part (a)

```
ggplot(starbucks_data, aes(x = carb, y = calories)) +  
  geom_point(alpha = 0.3) +  
  labs(x = "Carbohydrate Content (in grams)",  
       y = "Energy Content (in kilocalories)",  
       title = "Carbohydrate Content versus Energy Content  
of Starbucks Menu in 2011") +  
  geom_smooth(method = "lm", se = TRUE)
```

Carbohydrate Content versus Energy Content of Starbucks Menu in 2011



```
summary(lm(calories ~ carb, data = starbucks_data))
```

Call:

```
lm(formula = calories ~ carb, data = starbucks_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-151.962	-70.556	-0.636	54.908	179.444

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	146.0204	25.9186	5.634	2.93e-07 ***
carb	4.2971	0.5424	7.923	1.67e-11 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 78.26 on 75 degrees of freedom

Multiple R-squared: 0.4556, Adjusted R-squared: 0.4484

F-statistic: 62.77 on 1 and 75 DF, p-value: 1.673e-11

The estimated slope parameter is 4.2971 kilocalories per gram of carbohydrates. The interpretation of the slope parameter is:

for every gram increase of carbohydrates of a menu item, on average,

the energy content increases by 4.2971 kilocalories.

Part (b)

```
summary(lm(calories ~ fat + carb + fiber + protein, data = starbucks_data))
```

Call:

```
lm(formula = calories ~ fat + carb + fiber + protein, data = starbucks_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-24.236	-5.623	0.277	4.579	34.875

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.33360	4.14940	1.285	0.203
fat	8.95351	0.18172	49.270	<2e-16 ***
carb	3.84206	0.07799	49.265	<2e-16 ***
fiber	-0.02447	0.69953	-0.035	0.972
protein	3.99789	0.18370	21.763	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.56 on 72 degrees of freedom

Multiple R-squared: 0.9905, Adjusted R-squared: 0.99

F-statistic: 1874 on 4 and 72 DF, p-value: < 2.2e-16

The estimated slope parameter for `carb` is 3.84206. I would interpret the parameter as saying that for every gram increase of carbohydrates of a menu item, on average, the energy content increases by 3.84206 kilocalories assuming we hold the other factors (fat, fiber, and protein) constant.

Part (c)

I think that the estimator provided in part (b) is better. The model incorporates more treatment variables, and since it is highly likely that `fat`, `fiber`, and `protein` are correlated with `carbs`,

then they should have an effect on the net calories that a person gains from `carbs`. The body has to expend extra energy digesting a piece of food that is composed of multiple different macronutrients instead of just one. That extra energy decreases the net amount of energy that the body can extract from food, which lines up with the `carb` estimate being smaller.

Problem 5

Parts (a) and (b)

```
longbeach_clean <- longbeach$longbeach

longbeach_relevant <- longbeach_clean %>%
  filter(animal_type == "dog", intake_condition == "normal", outcome_type ==
    "adoption") %>%
  mutate(time_to_adoption = as.numeric(outcome_date - intake_date), age =
    as.numeric(intake_date - dob))

sum(longbeach_relevant$age < 0, na.rm = TRUE)
```

```
[1] 2
```

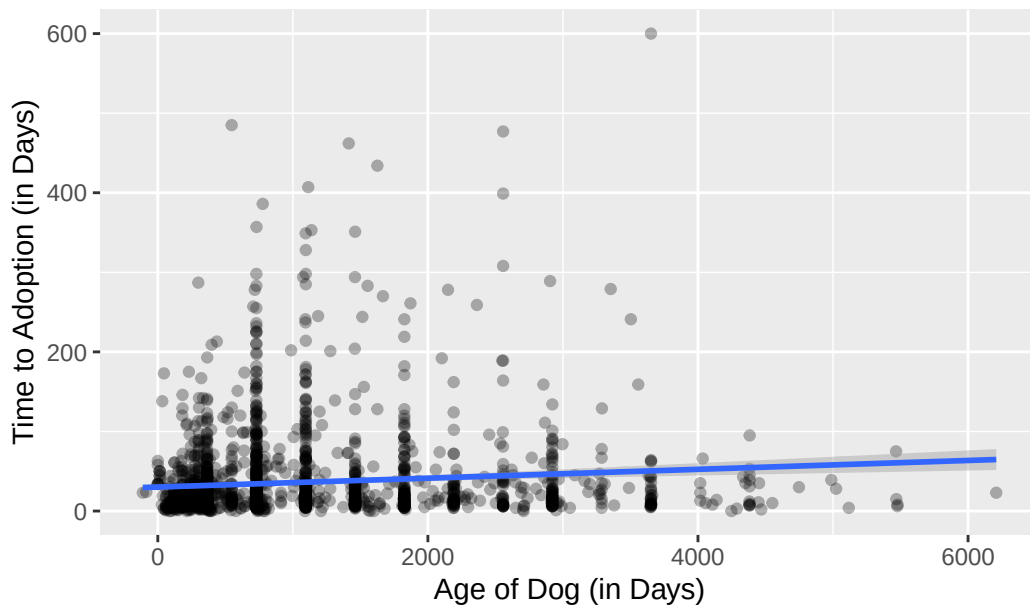
```
ggplot(data = longbeach_relevant, aes(x = age, y = time_to_adoption)) +
  geom_point(alpha = 0.3) +
  labs(x = "Age of Dog (in Days)",
       y = "Time to Adoption (in Days)",
       title = "Dog age on intake versus time to adoption") +
  geom_smooth(method = "lm", se = TRUE)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 24 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 24 rows containing missing values or values outside the scale range
(`geom_point()`).
```

Dog age on intake versus time to adoption



```
summary(longbeach_relevant)
```

animal_id	animal_name	animal_type	primary_color
Length :2052	Length :2052	Length :2052	Length :2052
N.unique :1954	N.unique :1555	N.unique : 1	N.unique : 30
N.blank : 0	N.blank : 0	N.blank : 0	N.blank : 0
Min.nchar: 7	Min.nchar: 2	Min.nchar: 3	Min.nchar: 3
Max.nchar: 7	Max.nchar: 18	Max.nchar: 3	Max.nchar: 13
	NAs : 23		

secondary_color	sex	dob	intake_date
Length :2052	Length :2052	Min. :2000-12-06	Min. :2017-01-01
N.unique : 18	N.unique : 4	1st Qu.:2016-10-24	1st Qu.:2020-01-27
N.blank : 0	N.blank : 0	Median :2020-03-25	Median :2022-09-11
Min.nchar: 3	Min.nchar: 4	Mean :2019-03-24	Mean :2022-01-09
Max.nchar: 13	Max.nchar: 8	3rd Qu.:2022-05-20	3rd Qu.:2023-11-25
NAs : 758		Max. :2024-07-20	Max. :2024-12-20
		NAs :24	

intake_condition	intake_type	intake_subtype	reason_for_intake
Length :2052	Length :2052	Length :2052	Length :2052
N.unique : 1	N.unique : 8	N.unique : 13	N.unique : 37
N.blank : 0	N.blank : 0	N.blank : 0	N.blank : 0

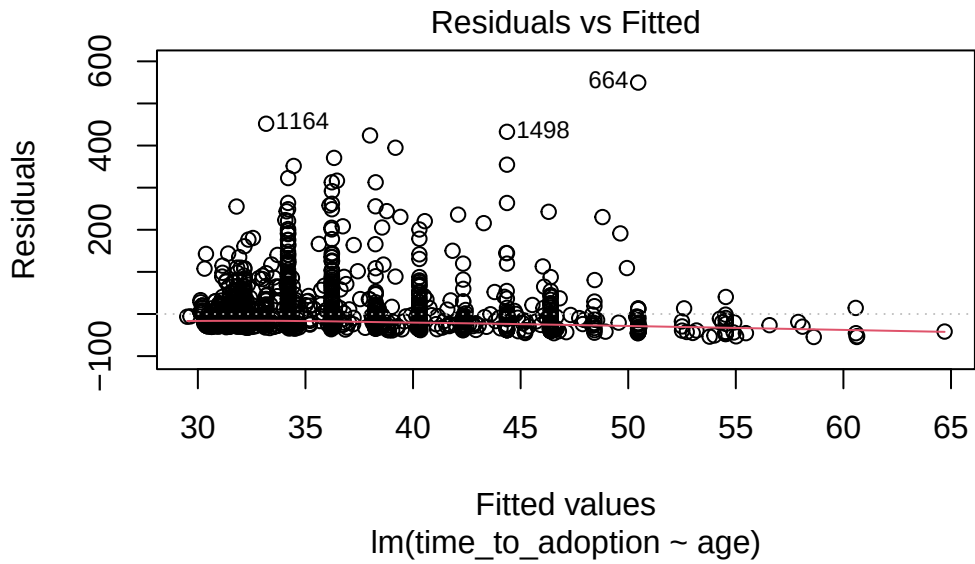
Min.nchar:	6	Min.nchar:	5	Min.nchar:	3	Min.nchar:	3
Max.nchar:	6	Max.nchar:	21	Max.nchar:	10	Max.nchar:	10
				NAs	: 37	NAs	:1765

outcome_date		crossing		jurisdiction		outcome_type	
Min.	:2017-01-08	Length	:2052	Length	:2052	Length	:2052
1st Qu.:	2020-02-27	N.unique	:1795	N.unique	: 16	N.unique	: 1
Median	:2022-10-10	N.blank	: 0	N.blank	: 0	N.blank	: 0
Mean	:2022-02-14	Min.nchar:	12	Min.nchar:	6	Min.nchar:	8
3rd Qu.:	2024-01-05	Max.nchar:	69	Max.nchar:	12	Max.nchar:	8
Max.	:2024-12-29						

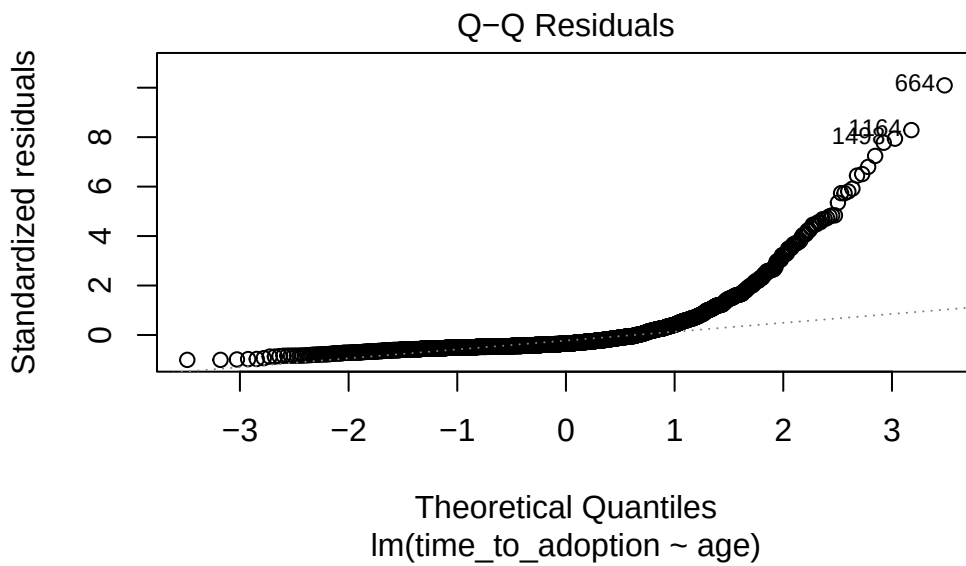
outcome_subtype		latitude		longitude		outcome_is_dead	
Length	:2052	Min.	:19.30	Min.	: -119.08	Mode	:logical
N.unique	: 12	1st Qu.:	33.78	1st Qu.:	-118.19	FALSE:	2052
N.blank	: 0	Median	:33.81	Median	: -118.18		
Min.nchar:	3	Mean	:33.81	Mean	: -118.15		
Max.nchar:	10	3rd Qu.:	33.85	3rd Qu.:	-118.15		
NAs	: 538	Max.	:35.30	Max.	: -99.01		

was_outcome_alive		geopoint		time_to_adoption		age	
Mode:logical		Length	:2052	Min.	: 0.00	Min.	: -111
TRUE:2052		N.unique	:1529	1st Qu.:	8.00	1st Qu.:	365
		N.blank	: 0	Median	: 16.00	Median	: 730
		Min.nchar:	20	Mean	: 35.59	Mean	:1020
		Max.nchar:	31	3rd Qu.:	37.00	3rd Qu.:	1461
				Max.	:600.00	Max.	:6209
						NAs	:24

```
regression <- lm(time_to_adoption ~ age, data = longbeach_relevant)
plot(regression, which = 1)
```



```
plot(regression, which = 2)
```



```
summary(regression)
```

Call:

```
lm(formula = time_to_adoption ~ age, data = longbeach_relevant)
```

Residuals:

Min	1Q	Median	3Q	Max
-54.62	-25.52	-19.16	0.84	549.54

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	30.126072	1.774937	16.973	< 2e-16 ***
age	0.005569	0.001272	4.378	1.26e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 54.55 on 2026 degrees of freedom

(24 observations deleted due to missingness)

Multiple R-squared: 0.009371, Adjusted R-squared: 0.008882

F-statistic: 19.17 on 1 and 2026 DF, p-value: 1.26e-05

There is some odd structure associated with this data. For one, there are two dogs that have a negative age. This does not provide confidence with the quality of the data. Secondly, we can see on the plot that there are regularly spaced spikes that are roughly 365 days spread apart. This suggests that when a new dog comes into the shelter, unless they happen to know precisely the date of birth of the dog, they must *approximate* the age. The highest resolution level that a staff member could reasonably guess is the number of years. This makes the dog have an age that is a clean, easy number.

Regarding the linear regression, the estimated slope parameter is 0.005569 adoption days per age days. This means that, on average, for every day that the dog increases in age, the time it takes for the dog to be adopted increases by 0.005569 days.

Part (c)

When looking at both the Q-Q plot and the Residuals vs Fitted Plot, they do **not** suggest normality.

Tackling each assumption turn by turn, it's safe to assume that the errors are **not** independent **nor** do they have mean zero. If they did, then we would see roughly equal amounts of data appear above and below the regression line. This is clearly not the case for our data.

The variance of the errors are likely finite, but as was previously mentioned, the data is **not** evenly spread above and below the regression line. The data is *much* more spread out or diffuse above the regression line than it is below it.

Combining the two previous observations, it is safe to say that the errors are **not** normally distributed.

Part (d)

No, I would say that this is not a good model to use.